

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Scenario-Based Questions

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA0510

COURSE OUTCOME COVERED

CO2: Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)

Assessment Tool Weightage: 50%

QUESTIONS: 5

Total Marks 50

Duration :60 Minutes

Annexure A

Scenario-Based Questions

Code of Conduct:

I Premarubha N (192525020) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Premarubha N

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks (100)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario	
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues	
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts	
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided	
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand	

Scenario-Based Questions

1. Low Contrast Image

A grayscale image appears dull with very little difference between object and background intensities.

- (a) Clearly interpret the scenario and explain the cause of low contrast.
- (b) Identify all key issues affecting image visibility.
- (c) Apply an appropriate enhancement technique to address the problem.
- (d) Justify your choice with logical reasoning.
- (e) Present your explanation in a clear and structured manner.

2. Uneven Illumination

An image captured under non-uniform lighting shows bright and dark patches.

- (a) Interpret the scenario and explain the impact of uneven illumination.
- (b) Identify key factors causing intensity variation.
- (c) Apply a suitable enhancement method to normalize illumination.
- (d) Justify your selected approach with reasoning.
- (e) Provide a clear and well-structured explanation.

3. Salt-and-Pepper Noise

A scanned document contains salt-and-pepper noise affecting readability.

- (a) Explain the scenario and characteristics of this noise.
- (b) Identify key issues impacting image quality.
- (c) Apply an appropriate filtering technique.
- (d) Justify why this method is more suitable than alternatives.
- (e) Present your response clearly and logically.

4. Blurred Edges

An image processed for noise removal appears overly smooth, causing edges to blur.

- (a) Interpret the scenario and explain why edges are blurred.
- (b) Identify key issues related to smoothing and edge loss.
- (c) Apply an appropriate technique to restore edges.
- (d) Justify your decision with clear reasoning.
- (e) Structure your explanation clearly.

5. High-Frequency Noise

A satellite image contains fine-grained noise affecting feature visibility.

- (a) Interpret the scenario and describe high-frequency noise.
- (b) Identify the key issues affecting image clarity.
- (c) Apply an appropriate frequency domain filtering method.
- (d) Justify the choice of filter with reasoning.
- (e) Present your answer in a clear and organized manner.



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ITA0510 – COMPUTER VISION
CO2 – ASSESMENT TOOL 2

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1. Low Contrast Image

(a) Clearly interpret the scenario and explain the cause of low contrast.

A grayscale image appears dull because the object and background have similar intensity values. This is caused by poor lighting, improper camera settings, or limited intensity range.

(b) Identify all key issues affecting image visibility.

- Poor visibility
- Loss of fine details
- Difficult object detection
- Reduced image quality

(c) Apply an appropriate enhancement technique.

Use Histogram Equalization.

Steps:

Compute histogram → Calculate CDF → Redistribute pixel intensities → Obtain enhanced image.

(d) Justify your choice.

It improves contrast by spreading intensity values over the full range, making hidden details visible.

(e) Present your explanation. Problem: Low contrast

Cause: Similar intensity values

Solution: Histogram Equalization

Result: Better visibility and improved image quality.

Conclusion: Histogram Equalization is the best method for enhancing low-contrast grayscale images.

2. Uneven Illumination

(a) Interpret the scenario and explain the impact.

Uneven lighting creates bright and dark regions, making image analysis difficult.

(b) Identify key factors.

- Non-uniform lighting
- Shadows
- Reflections
- Improper exposure

(c) Apply a suitable enhancement method.

Use Homomorphic Filtering to normalize illumination.

(d) Justify your approach.

It removes lighting variations while preserving important image details.

(e) Present your explanation.

Problem: Uneven brightness

Cause: Non-uniform illumination Solution: Homomorphic Filtering

Result: Uniform brightness and better image quality.

3. Salt-and-Pepper Noise

(a) Explain the scenario.

Random black and white pixels appear due to scanner or transmission errors.

(b) Identify key issues.

- Poor readability
- Distorted text
- Reduced image quality

(c) Apply an appropriate filtering technique. Use a Median Filter.

(d) Justify the method.

Median filtering removes impulse noise while preserving edges better than mean filtering.

(e) Present your explanation.

Problem: Salt-and-pepper noise

Solution: Median Filter

Result: Clean image with preserved edges.

4. Blurred Edges

(a) Interpret the scenario.

Excessive smoothing during noise removal blurs object boundaries.

(b) Identify key issues.

- Edge loss
- Reduced sharpness
- Poor feature detection

(c) Apply an appropriate technique.

Use Unsharp Masking (Image Sharpening).

(d) Justify your decision.

It restores edge sharpness without significantly increasing noise.

(e) Present your explanation.

Problem: Blurred edges

Cause: Over-smoothing

Solution: Unsharp Masking Result: Sharper image.

5. High-Frequency Noise

(a) Interpret the scenario.

Fine-grained noise reduces the visibility of satellite image features.

(b) Identify key issues.

- Reduced clarity
- Hidden details
- Poor feature extraction

(c) Apply a frequency domain filtering method. Use Gaussian Low-Pass Filter.

(d) Justify the filter.

It removes high-frequency noise smoothly while minimizing ringing effects.

(e) Present your explanation.

Problem: High-frequency noise

Solution: Gaussian Low-Pass Filter

Result: Improved clarity and feature visibility.